



Matrices · Mark Scheme

Core Pure 1 Chapter 6: Matrices. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Matrix arithmetic

- 1 Correct method, e.g. $\begin{pmatrix} 4-3 & -2-15 \\ 6+6 & 8-0 \end{pmatrix} M1; = \begin{pmatrix} 1 & -17 \\ 12 & 8 \end{pmatrix} A1.$
- 2 $\mathbf{AB} = \begin{pmatrix} 4 & 10 \\ -5 & 15 \end{pmatrix} M1 A1; \mathbf{BA} = \begin{pmatrix} 17 & 19 \\ -4 & 2 \end{pmatrix}$, so $\mathbf{AB} \neq \mathbf{BA}$: matrix multiplication is not commutative A1.
- 3 $\mathbf{CD}$ is 2×2 , $\mathbf{DC}$ is 3×3 B1; row-by-column method M1; $\mathbf{CD} = \begin{pmatrix} 4 & 9 \\ -1 & 0 \end{pmatrix} A1.$
- 4 (a) $\mathbf{A}^2 = \begin{pmatrix} a^2-2 & 2a+6 \\ -a-3 & 7 \end{pmatrix} M1; 2a+6=8 \Rightarrow a=1$ (checks with the other entries) A1. (b) $\begin{pmatrix} -1 & 8 \\ -4 & 7 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ -4 & 12 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} M1; = \mathbf{0} A1.$
- 5(a) $\mathbf{A}^2 = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix} B1; \mathbf{A}^3 = \begin{pmatrix} 8 & 19 \\ 0 & 27 \end{pmatrix} B1.$
- 5(b) $5\mathbf{A} - 6\mathbf{I} = \begin{pmatrix} 10-6 & 5 \\ 0 & 15-6 \end{pmatrix} M1; = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix} = \mathbf{A}^2 A1.$
- 5(c) $\mathbf{A}^4 = (5\mathbf{A} - 6\mathbf{I})^2 = 25\mathbf{A}^2 - 60\mathbf{A} + 36\mathbf{I} M1; = 25(5\mathbf{A} - 6\mathbf{I}) - 60\mathbf{A} + 36\mathbf{I} M1; = 65\mathbf{A} - 114\mathbf{I} A1.$

Section B – 2×2 determinants and inverses

- 6 $\det = 5 \times 2 - 3 \times 4 = -2 M1$; non-zero, so non-singular A1.
- 7 $\det = 15 - 8 = 7 B1$; swap the leading diagonal, change signs of the other M1; $\frac{1}{7} \begin{pmatrix} 5 & -4 \\ -2 & 3 \end{pmatrix} A1.$
- 8 $k(k+1) - 12 = 0 M1; k^2 + k - 12 = (k+4)(k-3) = 0 M1; k=3$ or $k=-4 A1.$
- 9 $\det \mathbf{A} = 1, \mathbf{A}^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} B1; \mathbf{B} = \mathbf{A}^{-1}(\mathbf{AB})$ (pre-multiply) M1; $= \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix} M1; = \begin{pmatrix} 0 & -1 \\ 1 & 4 \end{pmatrix} A1.$
- 10(a) $\det \mathbf{M} = k(k-4) + 7 = k^2 - 4k + 7 M1; = (k-2)^2 + 3 M1; \geq 3 > 0$ for all k , so $\mathbf{M}$ is non-singular A1.
- 10(b) $\mathbf{M}^{-1} = \frac{1}{k^2 - 4k + 7} \begin{pmatrix} k-4 & -7 \\ 1 & k \end{pmatrix} M1 A1.$
- 10(c) $\mathbf{M}^{-1} = \frac{1}{3} \begin{pmatrix} -2 & -7 \\ 1 & 2 \end{pmatrix} B1; \mathbf{v} = \mathbf{M}^{-1} \begin{pmatrix} 16 \\ -5 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -32+35 \\ 16-10 \end{pmatrix} M1; = \begin{pmatrix} 1 \\ 2 \end{pmatrix} A1.$

Section C – 3×3 determinants and inverses

- 11 $2 \begin{vmatrix} 1 & -1 \\ 4 & 2 \end{vmatrix} - 0 + 1 \begin{vmatrix} 3 & 1 \\ 0 & 4 \end{vmatrix} = 2(6) + 12 M1; = 24 A1.$
- 12 $1(2k-1) - 2(0-3) + k(0-3k) M1; -3k^2 + 2k + 5 = 0 \Rightarrow (3k-5)(k+1) = 0 M1; k = \frac{5}{3}$ or $k = -1 A1.$
- 13 Minors with the sign pattern M1; cofactors $\begin{pmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{pmatrix} A1; \det \mathbf{A} = 1(1) + 2(6) + 0(-2) = 13 A1.$



14 $\det \mathbf{B} = 1(2) - 1(-1) + 0 = 3$ *M1*; cofactors $\begin{pmatrix} 2 & 1 & -1 \\ -2 & 2 & 1 \\ 1 & -1 & 1 \end{pmatrix}$ *M1*;

transpose to the adjugate *M1*; $\mathbf{B}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & -1 \\ -1 & 1 & 1 \end{pmatrix}$ *A1*.

15(a) $1(1-0) - a(0-a^2) + 0$ *M1*; $= 1 + a^3$ *A1*.

15(b) $a = -1$ *B1*.

15(c) At least 4 correct cofactors *M1*; cofactors $\begin{pmatrix} 1 & a^2 & -a \\ -a & 1 & a^2 \\ a^2 & -a & 1 \end{pmatrix}$ *A1*; transpose and divide by the determinant *M1*;

$\mathbf{M}^{-1} = \frac{1}{1+a^3} \begin{pmatrix} 1 & -a & a^2 \\ a^2 & 1 & -a \\ -a & a^2 & 1 \end{pmatrix}$ *A1*.

Section D – Systems of linear equations

16 $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 5 & -4 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 10 \\ 9 \end{pmatrix}$ *M1*; $x = 2, y = 1$ *A1*.

17 $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{N}^{-1} \begin{pmatrix} 9 \\ 0 \\ 1 \end{pmatrix}$ *M1*; $= \frac{1}{7} \begin{pmatrix} 9+0-2 \\ -18+0+4 \\ 18-0+3 \end{pmatrix}$ *M1*; $x = 1, y = -2, z = 3$ *A1*.

18 No unique solution when the determinant is zero *M1*; $1(2k-3) - 2(4-3) + 1(2-k) = k-3$ *A1*; $k = 3$ *A1*.

19 (a) $\det = 1(-4-2) - 1(4-6) + 1(1+3) = 0$ *M1*; so no unique point of intersection *A1*. (b) Row 3 = $2 \times$ row 1 + row 2 on the left, but $2(2) + 1 = 5 \neq 6$, so inconsistent *M1*; no two normals are parallel, so the planes form a triangular prism *A1*.

20(a) $1(-3-2k) - 1(6-k) + 1(4+1)$ *M1*; $= -k-4$ *A1*.

20(b) Correct method, e.g. eliminating one variable or using the inverse matrix *M1*; at least two values correct *A1*; $x = 1, y = 2, z = 3$ *A1*.

20(c) Determinant is 0 and, e.g., $5 \times (\text{eqn 1}) - 3 \times (\text{eqn 3})$ gives $2x - y - 4z = -12$ *M1*; but eqn 2 is $2x - y - 4z = 3$, so the equations are inconsistent *A1*; no two planes are parallel, so the three planes form a triangular prism *B1*.

Roar-level challenge

21 $\mathbf{A}^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ *M1*; $= \mathbf{A} + 2\mathbf{I}$ *A1*; $\mathbf{A}^2 - \mathbf{A} = 2\mathbf{I} \Rightarrow \mathbf{A}(\mathbf{A} - \mathbf{I}) = 2\mathbf{I}$ *M1*; $\mathbf{A}^{-1} = \frac{1}{2}(\mathbf{A} - \mathbf{I})$ *A1*; $= \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$ *A1*.